

Analog Buck-Boost (LTC3440, 3.9 V): Working Document

Digital Audio Player • Rev 8

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1. Summary

Field	Value
Manufacturer / MPN	Analog Devices, LTC3440EMS#TRPBF
Package	10-lead MSOP (MS)
Function	Synchronous buck-boost, SYS → 3.9 V intermediate analog rail

2. Design Requirements

Requirement	Value
Input	SYS, 3.0 to 4.2 V
Output	3.9 V
Load	Analog chain only: LT3042 (+3.3 V) and LT3462A inverter → LT3093 (−3.3 V)
Output current	≈ 253 mA peak (106 mA LT3042 + 147 mA LT3462A input, simulated); 250 mA design current, exceeded by about 1%
Fault ceiling	≈ 0.3 A (LT3042 at 125 mA + LT3462A at its switch limit)
Switching frequency	1.2 MHz, fixed (Burst Mode off)
Output ripple target	0.5% (19.5 mV pp)
Soft-start	Output up in ≈ 6 ms, MCU-enabled
Source resistance	≈ 0.21 Ω
Constraints	Low noise · Small footprint

3. Pin Configuration

Pin	Name	Connection
1	RT	R_T to GND
2	MODE/SYNC	GND (fixed frequency)
3	SW1	L
4	SW2	L
5	GND	GND
6	VOUT	+3V9, C_{OUT} , R1
7	VIN	SYS, C_{IN}
8	SHDN/SS	R_{SS} from GPIO, C_{SS} to GND
9	FB	R1 / R2 midpoint
10	VC	C_{P1} and R_Z to FB

4. External Components

Ref	Value / Rating	MPN	AL-PN
L	6.8 μ H $\pm$ 20%, I_{sat} 3.5 A (20%), 74.1 m Ω	Bourns SRP4030FA-6R8M	#1-076
C_{IN}	4.7 μ F 16 V X7R 0805	Murata GRM21BR71C475KE51L	#1-071
C_{OUT}	22 μ F 25 V X7S 1210	Murata GCM32EC71E226KE36L	#1-073
R1	220 k Ω 0.1% 0402	YAGEO RC0402BR-07220KL	#1-074
R2	100 k Ω 0.1% 0603	YAGEO RT0603BRD07100KL	#1-066
R_T	49.9 k Ω 0.1% 0402	YAGEO RT0402BRD0749K9L	#1-138
R_Z	15 k Ω 0.1% 0402	YAGEO RT0402BRD0715KL	#1-078
C_{P1}	1.5 nF $\pm$ 2% 50 V C0G 0402	Murata GRM1555C1H152GA01D	#1-077
R_{SS}	54.9 k Ω 0.1% 0402	YAGEO RT0402BRD0754K9L	#1-079
C_{SS}	0.1 μ F 50 V X7R 0402	Murata GCM155R71H104KE02D	#1-080
R_{PD}	1 M Ω , GPIO side of R_{SS}		

5. Equations and Results

Switching frequency and output voltage

$$R_T = \frac{6 \times 10^{10}}{1.2 \times 10^6} = 50 \text{ k}\Omega \rightarrow 49.9 \text{ k}\Omega, \quad f = \frac{6 \times 10^{10}}{49.9 \text{ k}\Omega} = 1.202 \text{ MHz}$$
$$V_{OUT} = 1.22 \left(1 + \frac{220}{100} \right) = 3.904 \text{ V} \quad (3.827 \text{ to } 3.981 \text{ V})$$

Source: [LTC3440 Product Page and Datasheet \(Analog Devices\)](#).

Inductor ripple (f in MHz; 250 mA design current)

$$\text{Ripple}_{\text{boost}} = \frac{V_{IN(MIN)} (V_{OUT} - V_{IN(MIN)})}{f I_{OUT} L V_{OUT}} = \frac{3.0 \times 0.9}{1.2 \times 0.25 \times 6.8 \times 3.9} = 0.34 \text{ (85 mA}_{pp}\text{)}$$
$$\text{Ripple}_{\text{buck}} = \frac{V_{OUT} (V_{IN(MAX)} - V_{OUT})}{f I_{OUT} L V_{IN(MAX)}} = 0.14 \text{ (34 mA}_{pp}\text{)}$$

Source: [LTC3440 Product Page and Datasheet \(Analog Devices\)](#), inductor selection.

Peak inductor current (3.0 V in, $\eta \approx 0.85$) and output capacitance (0.5% ripple)

$$I_{L, pk} = \frac{0.25 \times 3.9}{3.0 \times 0.85} + \frac{0.085}{2} = 0.42 \text{ A}$$
$$C_{OUT} \geq \frac{0.25 \times 0.9 \times 100}{0.5 \times 3.9^2 \times 1.2 \times 10^6} = 2.47 \mu\text{F}$$

Source: [LTC3440 Product Page and Datasheet \(Analog Devices\)](#).

Soft-start

$$\tau = R_{SS}C_{SS} = 5.49 \text{ ms}, \quad t_x = \tau \ln \frac{V_{GPIO}}{V_{GPIO} - V_x} \Rightarrow \text{enable (1.0 V) : 2.0 ms, ramp complete (2.2 V) : 6.0 ms}$$

Source: [LTC3440 Product Page and Datasheet \(Analog Devices\)](#), SHDN/SS pin.

Four-switch region

$$V_{IN} = V_{OUT}(1 - 150 \text{ ns} \cdot f) \text{ to } \frac{V_{OUT}}{1 - 150 \text{ ns} \cdot f} = 3.20 \text{ to } 4.76 \text{ V}$$

Source: [LTC3440 Product Page and Datasheet \(Analog Devices\)](#).

Case	LT3042	LT3462A input (simulated)	LTC3440 load
Quiescent	35 mA	$\approx 38 \text{ mA}$	$\approx 73 \text{ mA}$
Full scale, 23.8 Ω	106 mA	$\approx 147 \text{ mA}$	$\approx 253 \text{ mA}$
Fault (LDOs / inverter in limit)	125 mA	$\approx 0.18 \text{ A}$	$\approx 0.3 \text{ A}$

6. Important Values

Item	Value
Switch current limit	2.7 A typ (2.5 to 2.9 A)
Simulated output ripple	7.7 mV pp (target 19.5 mV)
Load step 150 $\leftrightarrow$ 250 mA	$\pm 90 \text{ mV}$, no ringing, 0.3 ms recovery
Loop crossover (implied)	$\approx 8 \text{ kHz}$
Minimum rail during step	3.74 V
ESL spikes (0.5 nH, simulated)	$\approx 0.3 \text{ V}$; layout item
Fault ceiling	$\approx 0.3 \text{ A}$: inductor peak $\approx 0.5 \text{ A}$ vs $3.5 \text{ A } I_{sat}$; switch 2.5 A min

7. Open Problems

None.

8. Revision Log

Rev	Date	Change
0	2026-09-21	Initial design, calculations and simulation.
1	2026-09-21	Schematic captured; VC-to-FB short corrected.
2	2026-09-21	Condensed to essentials.
3	2026-09-22	Load updated to 0.6 V RMS maximum output; 250 mA fault ceiling.
4	2026-09-23	Negative branch now the LT3462A inverter: peak $\approx$ 249 mA, fault $\approx$ 0.3 A.
5	2026-09-25	Reformatted; linked contents and sources.
6	2026-09-25	LT3462A efficiency open problem closed: simulation gives 67.7% at 98 mA ($\approx$ 147 mA input), below the 151 mA design assumption.
7	2026-09-27	Load updated from Power Budget Rev 6: LT3042 peak 106 mA, total $\approx$ 253 mA.
8	2026-10-08	R_{f} 50 k Ω 0603 (#1-075) $\rightarrow$ 49.9 k Ω 0402 RT0402BRD0749K9L (#1-138); f = 1.202 MHz.

9. Sources

1. [LTC3440 Product Page and Datasheet \(Analog Devices\)](#)